

Econometria

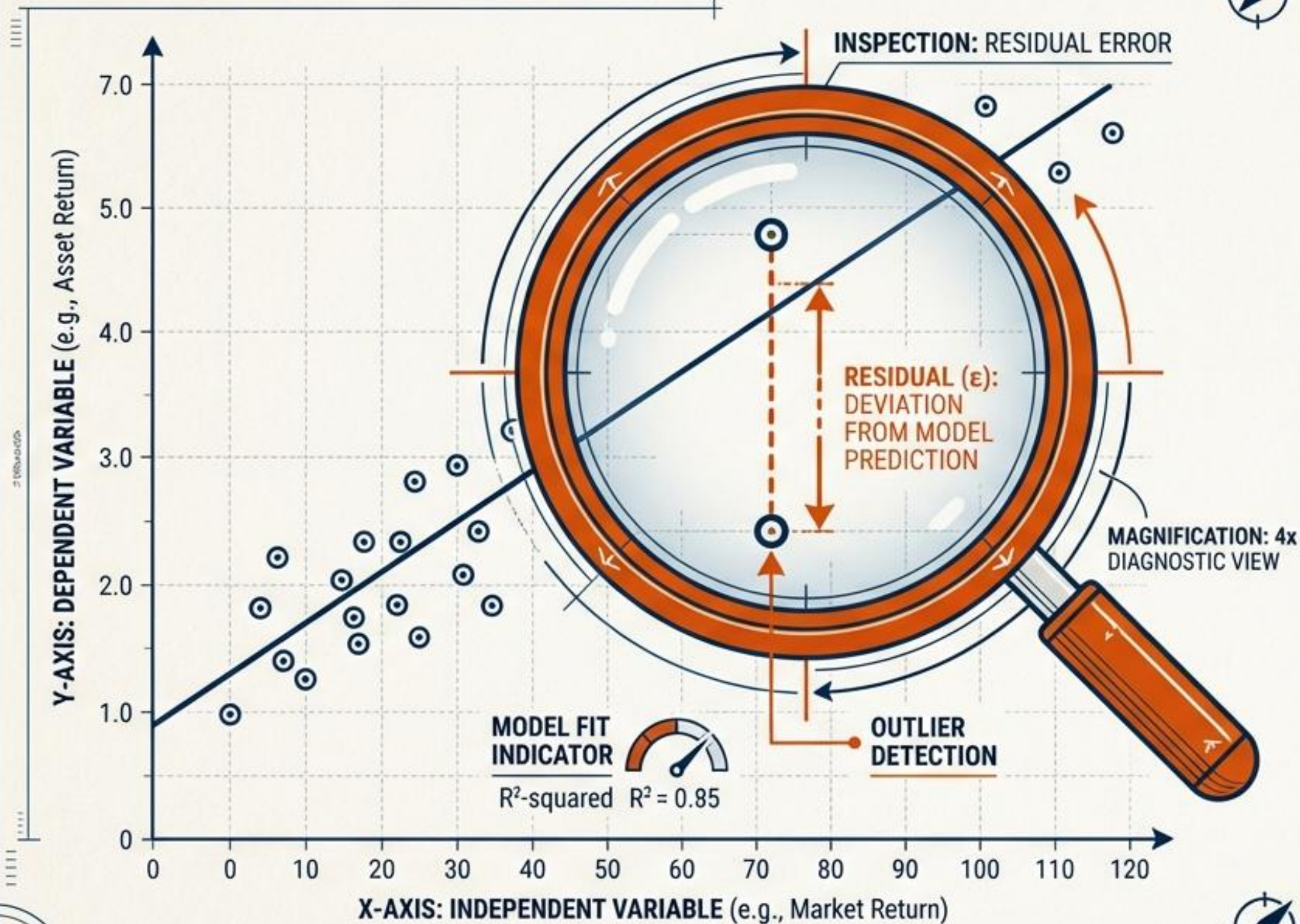
Aula 4: Desenvolvimento Adicionais - Regressão Linear

Prof. José Américo



<https://econometrics-lab.streamlit.app/>

EXHIBIT A: REGRESSION ANALYSIS DIAGNOSTIC



HELVETICA NOW DISPLAY

Diagnóstico e Validação de Modelos Econométricos

Um Guia Executivo para “Check-ups” de Saúde em Regressões Lineares e Tomada de Decisão Robusta.

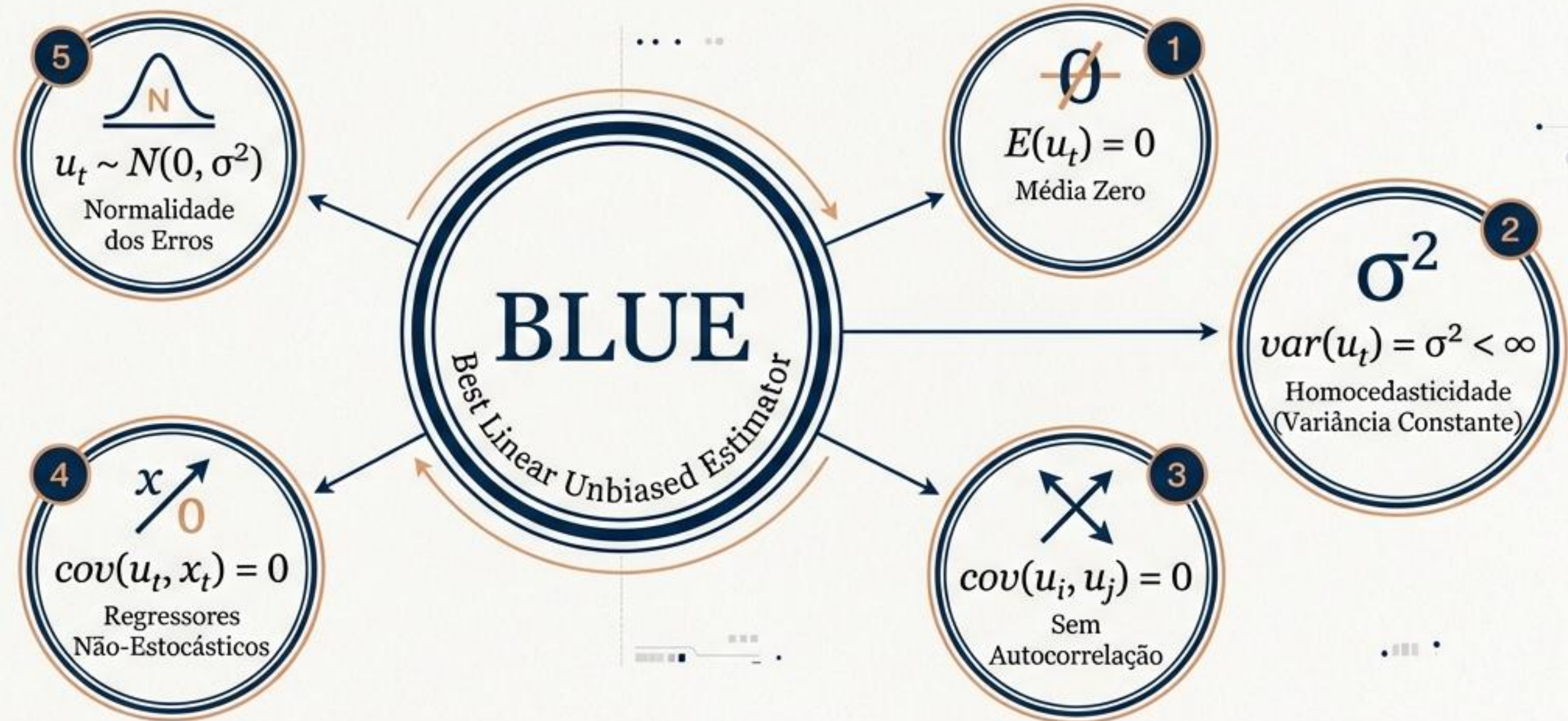
KEY OBJECTIVES:

1. Identify Model Misspecification;
2. Assess Heteroskedasticity & Autocorrelation;
3. Ensure Robustness for Forecasting & Policy.

Baseado no Capítulo 5 de *Introductory Econometrics for Finance*.

O Padrão Ouro: O Estimador BLUE

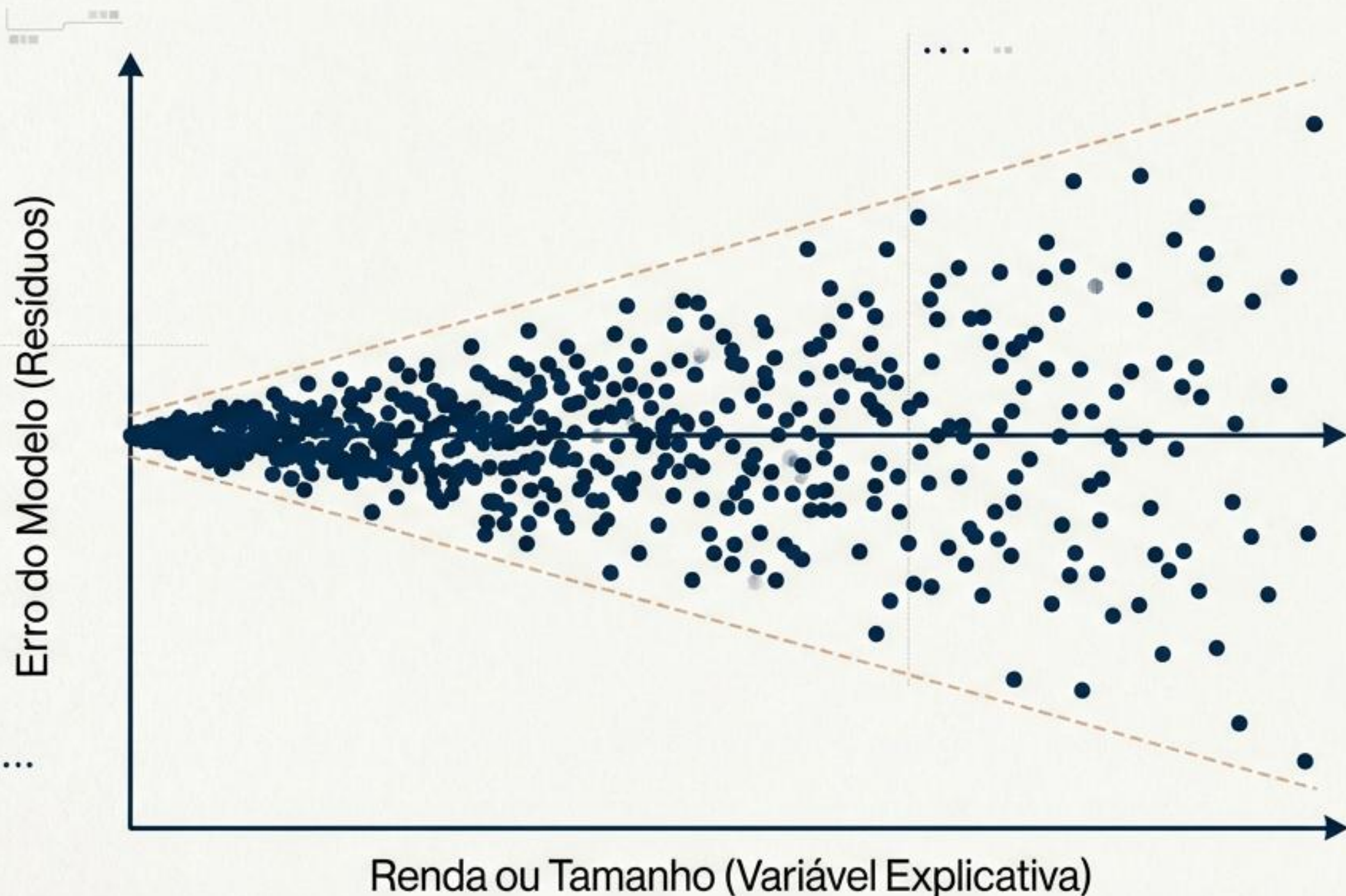
Para confiar no diagnóstico, o paciente deve ser estável.



Contexto: Por que isso importa? Se essas premissas se sustentam, seu modelo é a estimativa mais precisa possível. Se falharem, suas previsões financeiras podem estar enviesadas ou seus cálculos de risco (erros padrão) estarão errados.

Diagnóstico 1: Variância dos Erros (Heterocedasticidade)

Quando a incerteza aumenta com o tamanho da aposta.



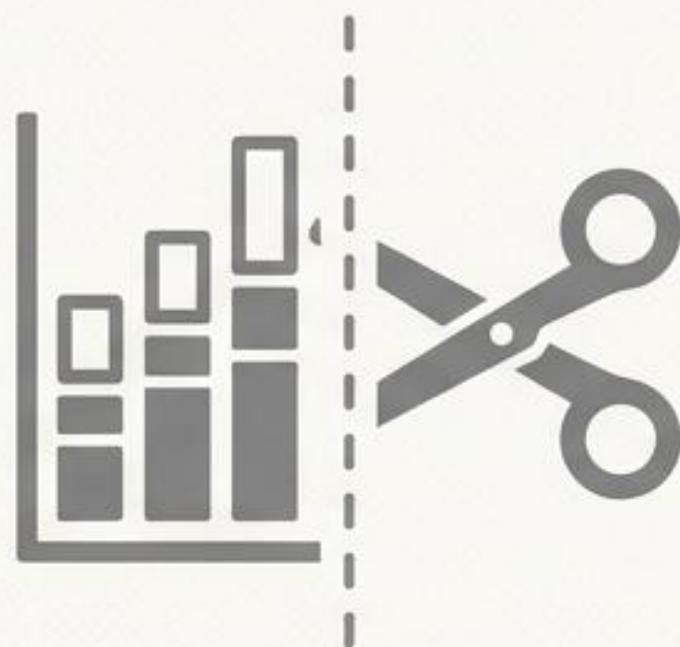
****Sintoma Visual:**
O erro não é constante.

Exemplo imobiliário:
Prever o preço de um apartamento pequeno é fácil (erro baixo).
Prever o preço de uma mansão é difícil (erro alto).

Detectando a Heterocedasticidade: O Teste de White

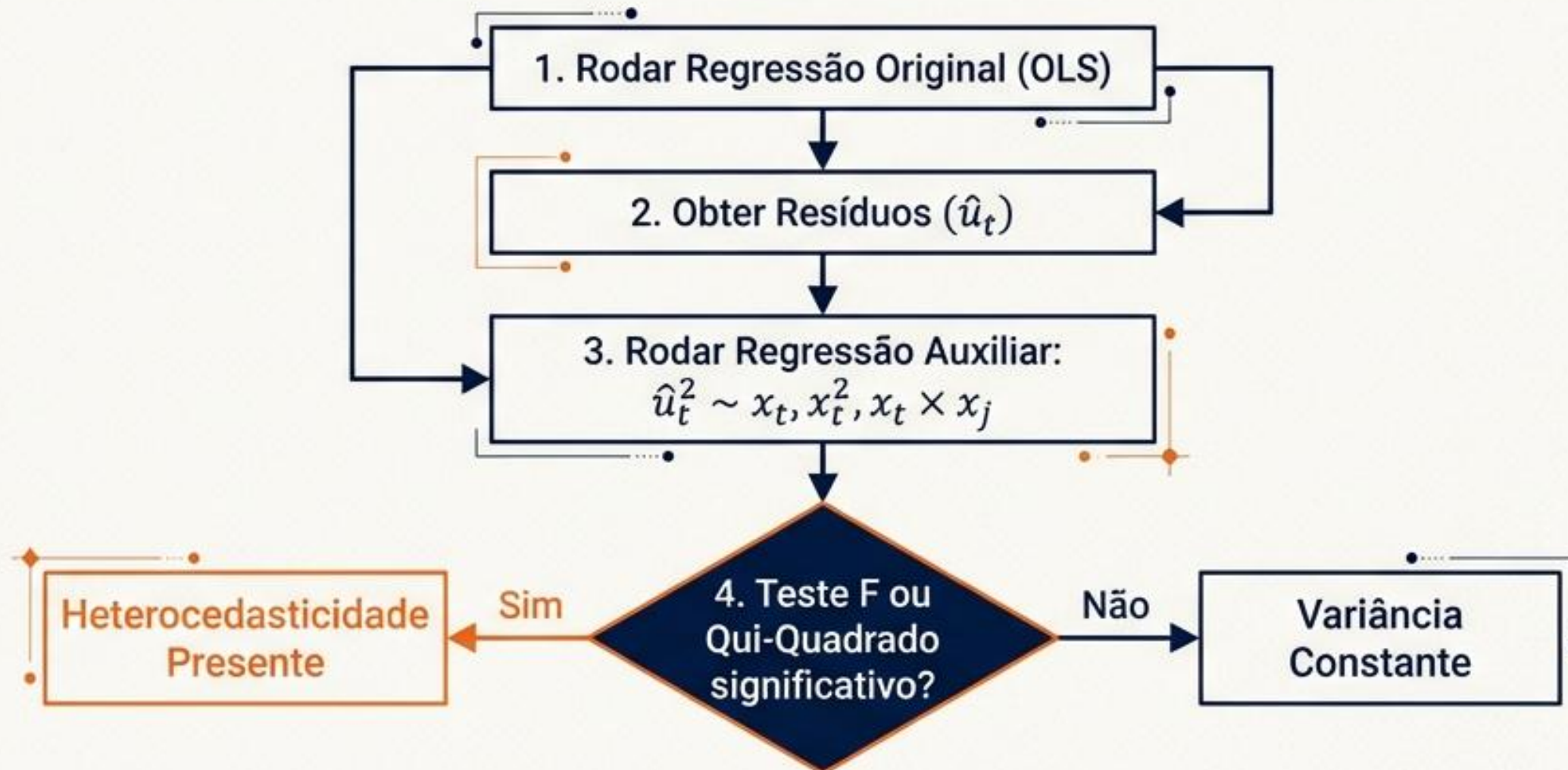
Teste Goldfeld-Quandt

Requer dividir a amostra arbitrariamente. Menos eficiente e exige premissas rígidas.



Teste de White (Padrão da Indústria)

Verifica se a variância do erro (u^2) está correlacionada com as variáveis explicativas (x) ou seus quadrados (x^2).



Contexto: O teste de White é o padrão moderno devido à sua robustez e não requer premissas sobre a distribuição dos erros, tornando-o mais flexível que alternativas antigas.

Heterocedasticidade: Impacto e Tratamento

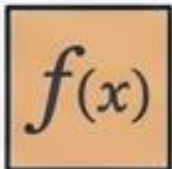


Diagnóstico - Impacto no Negócio

O Problema: A linha de tendência (coeficientes) ainda está correta, MAS a avaliação de risco (Erros Padrão) está errada.

Consequência: Você pode acreditar que uma variável é estatisticamente significativa quando ela não é. Testes-t e Intervalos de Confiança tornam-se inválidos.

Prescrição - Tratamento



1

Transformação

Usar Logaritmos ($\ln$). Comprime escalas e reduz a variância dos dados brutos.



2

GLS

Mínimos Quadrados Generalizados. Reponderar os dados, dividindo pela variável que causa a variância.



3

A Solução Prática (Star Pick)

Erros Padrão Robustos de White (White's SE)

Ajuste automático disponível em softwares modernos que corrige o viés nos testes de hipótese.

4- Homocedasticidade dos Resíduos

TESTE ARCH

View/Residual Diagnostics/Heteroskedasticity Test/ARCH

H0: Não há heterocedasticidade nos resíduos

Dependent Variable: CRED
Method: Least Squares
Date: 03/10/23 Time: 09:47
Sample (adjusted): 2006M04 2016M12
Included observations: 129 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	81.49080	5.988192	13.60858	0.0000
NPL(-1)	-7.367373	1.398951	-5.266354	0.0000
IR(-1)	-1.079624	0.258865	-4.170611	0.0001

R-squared	0.242342	Mean dependent var	43.77713
Adjusted R-squared	0.230316	S.D. dependent var	7.489106
S.E. of regression	6.570317	Akaike info criterion	6.625982
Sum squared resid	5439.302	Schwarz criterion	6.692490
Log likelihood	-424.3759	Hannan-Quinn criter.	6.653006
F-statistic	20.15097	Durbin-Watson stat	0.017075
Prob(F-statistic)	0.000000		

Heteroskedasticity Test: ARCH

F-statistic	2124.959	Prob. F(1,126)	0.0000
Obs*R-squared	120.8351	Prob. Chi-Square(1)	0.0000

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 03/16/23 Time: 16:03
Sample (adjusted): 2006M05 2016M12
Included observations: 128 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.293605	1.378638	0.938321	0.3499
RESID^2(-1)	0.981646	0.021295	46.09727	0.0000

R-squared	0.944024	Mean dependent var	42.05722
Adjusted R-squared	0.943580	S.D. dependent var	50.37751
S.E. of regression	11.96616	Akaike info criterion	7.817544
Sum squared resid	18041.81	Schwarz criterion	7.862107
Log likelihood	-498.3228	Hannan-Quinn criter.	7.835650
F-statistic	2124.959	Durbin-Watson stat	2.255253
Prob(F-statistic)	0.000000		

Dependent Variable: CRED
Method: Least Squares
Date: 03/10/23 Time: 09:51
Sample (adjusted): 2006M04 2016M12
Included observations: 129 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	31.13936	1.898327	16.40358	0.0000
NPL(-1)	-0.457424	0.385295	-1.187203	0.2374
IR(-1)	-0.737425	0.065386	-11.27798	0.0000
GDP(-1)	5.37E-06	1.24E-07	43.34862	0.0000

R-squared	0.952743	Mean dependent var	43.77713
Adjusted R-squared	0.951609	S.D. dependent var	7.489106
S.E. of regression	1.647447	Akaike info criterion	3.866848
Sum squared resid	339.2604	Schwarz criterion	3.955524
Log likelihood	-245.4117	Hannan-Quinn criter.	3.902879
F-statistic	840.0429	Durbin-Watson stat	0.052351
Prob(F-statistic)	0.000000		

Heteroskedasticity Test: ARCH

F-statistic	480.0847	Prob. F(1,126)	0.0000
Obs*R-squared	101.3899	Prob. Chi-Square(1)	0.0000

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 03/16/23 Time: 16:06
Sample (adjusted): 2006M05 2016M12
Included observations: 128 after adjustments

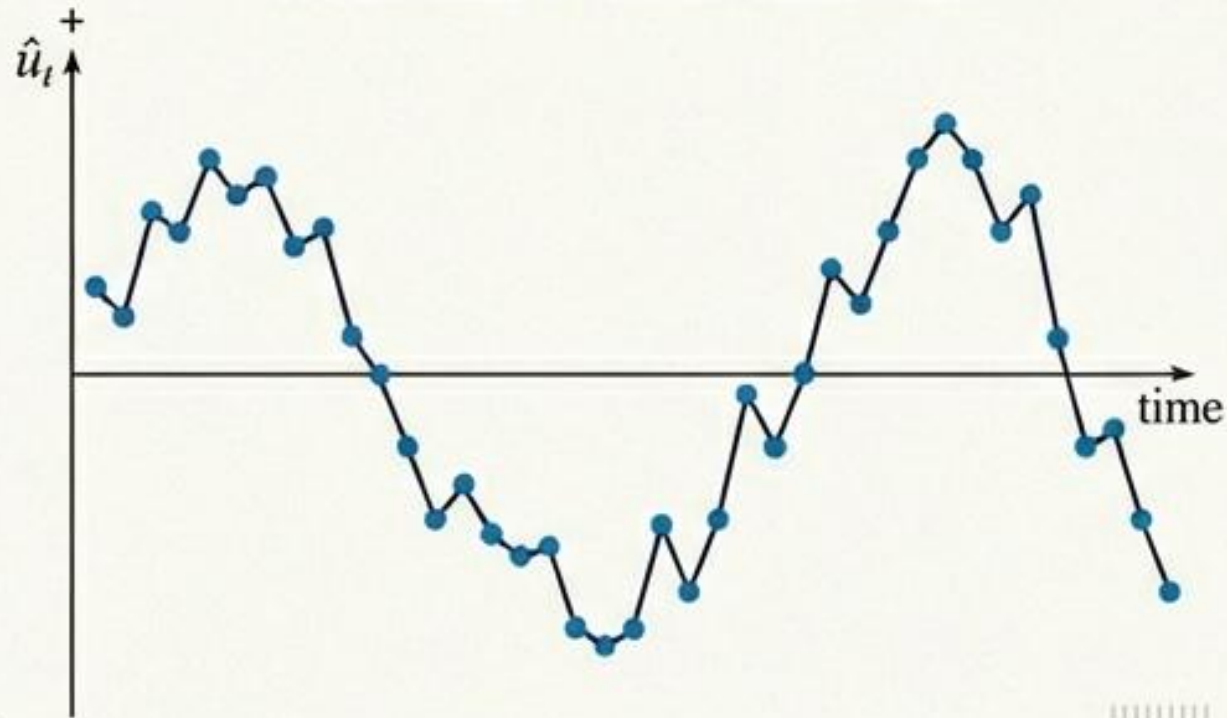
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.285367	0.147446	1.935403	0.0552
RESID^2(-1)	0.913553	0.041694	21.91084	0.0000

R-squared	0.792108	Mean dependent var	2.641631
Adjusted R-squared	0.790458	S.D. dependent var	2.493174
S.E. of regression	1.141268	Akaike info criterion	3.117659
Sum squared resid	164.1141	Schwarz criterion	3.162222
Log likelihood	-197.5302	Hannan-Quinn criter.	3.135765
F-statistic	480.0847	Durbin-Watson stat	1.298187
Prob(F-statistic)	0.000000		

Diagnóstico 2: Padrões Temporais (Autocorrelação)

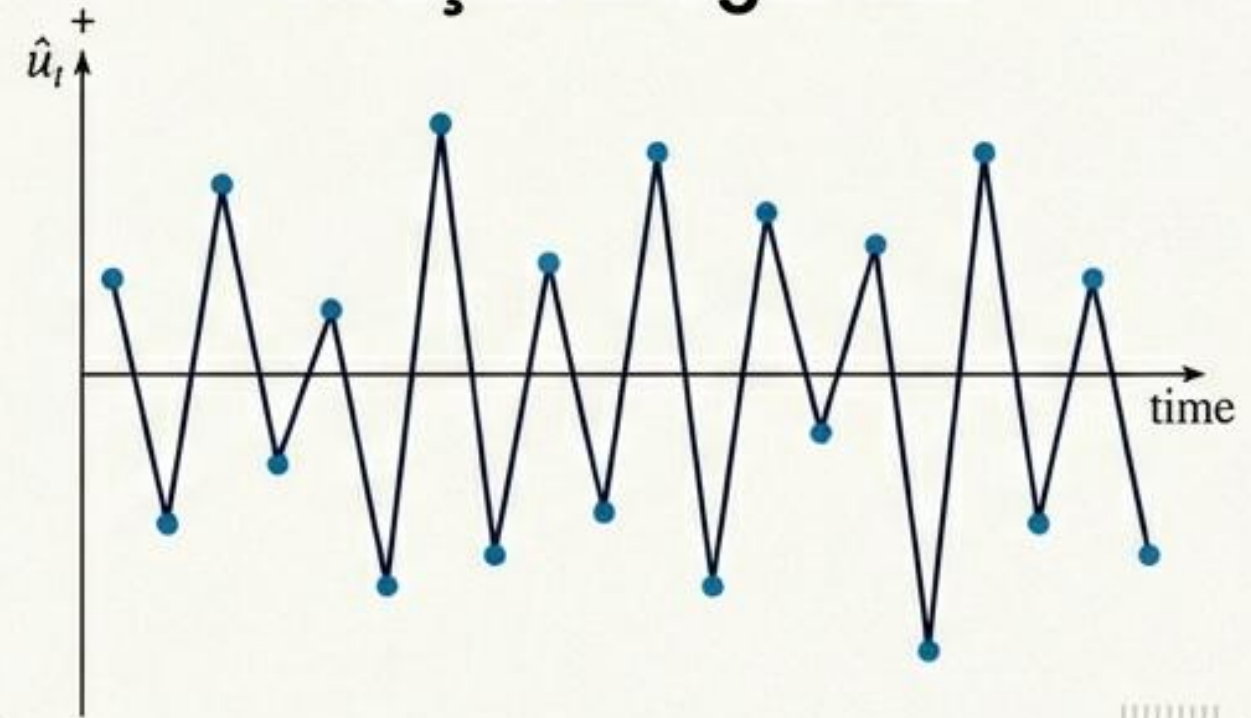
A memória do mercado: quando o erro de hoje prevê o erro de amanhã.

Inércia / Tendência



O erro de ontem “arrasta” o erro de hoje.

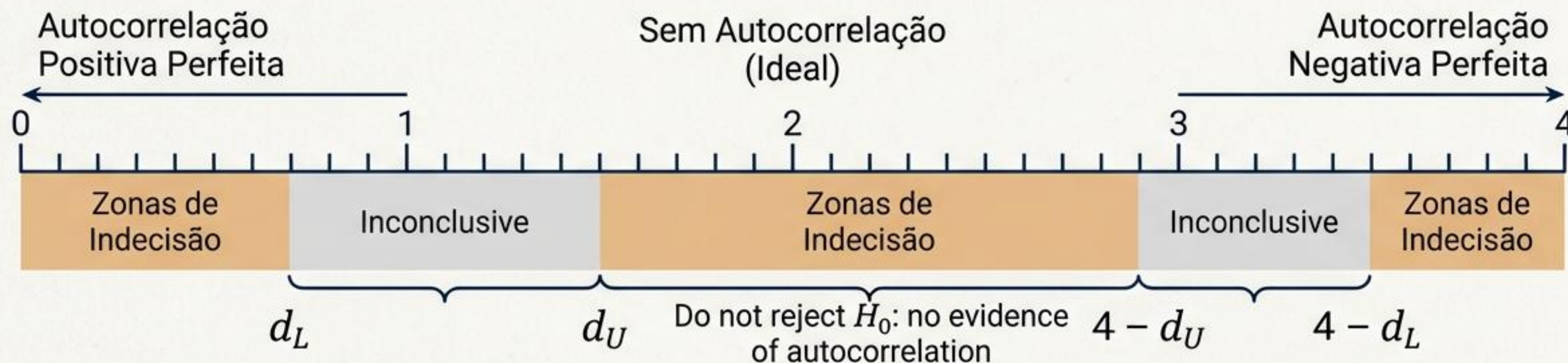
Reação Exagerada



Oscilação rápida e alternada.

Insight: Se o seu gráfico de resíduos tem um padrão visível, seu modelo deixou informação (dinheiro) na mesa.

Ferramentas de Detecção: DW vs. Breusch-Godfrey



Durbin-Watson	Breusch-Godfrey (LM Test)
• Limitado. • Só detecta relação imediata (t com $t - 1$). • Falha se houver lags da variável dependente.	• Padrão Ouro. • Mais flexível: detecta padrões de longo prazo (múltiplos lags). • Funciona com regressores estocásticos.

Autocorrelação: O Perigo da Falsa Precisão

The Trap



O modelo parece fantástico, mas é uma miragem. O OLS subestima a variância real, inflando artificialmente o R^2 e os testes-t. O modelo falhará na previsão real.

The Cure



Procedimento Cochrane-Orcutt:



- GLS Iterativo. Estima a correlação (ρ) e transforma os dados para remover o padrão.



Erros Padrão HAC (Newey-West):



- Ajuste robusto para Heterocedasticidade e Autocorrelação simultaneamente.



Modelagem Dinâmica:



- Adicionar *lags* (defasagens) para capturar a estrutura de tempo explicitamente.

5- Autocorrelação Serial dos Resíduos

TESTE LM

View/Residual Diagnostics/Serial Correlation LM Test

H0: Não há autocorrelação serial nos resíduos

Dependent Variable: CRED Method: Least Squares Date: 03/10/23 Time: 09:47 Sample (adjusted): 2006M04 2016M12 Included observations: 129 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	81.49080	5.988192	13.60858	0.0000
NPL(-1)	-7.367373	1.398951	-5.266354	0.0000
IR(-1)	-1.079624	0.258865	-4.170611	0.0001
R-squared	0.242342	Mean dependent var	43.77713	
Adjusted R-squared	0.230316	S.D. dependent var	7.489106	
S.E. of regression	6.570317	Akaike info criterion	6.625982	
Sum squared resid	5439.302	Schwarz criterion	6.692490	
Log likelihood	-424.3759	Hannan-Quinn criter.	6.653006	
F-statistic	20.15097	Durbin-Watson stat	0.017075	
Prob(F-statistic)	0.000000			

Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 2 lags

F-statistic	2293.357	Prob. F(2,124)	0.0000
Obs*R-squared	125.6043	Prob. Chi-Square(2)	0.0000

Test Equation:
Dependent Variable: RESID
Method: Least Squares
Date: 03/16/23 Time: 16:09
Sample: 2006M04 2016M12
Included observations: 129
Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.645006	0.991632	1.658888	0.0997
NPL(-1)	-0.404934	0.232325	-1.742962	0.0838
IR(-1)	-0.013075	0.042341	-0.308789	0.7580
RESID(-1)	0.883540	0.090010	9.815971	0.0000
RESID(-2)	0.118220	0.091422	1.293127	0.1984
R-squared	0.973677	Mean dependent var	-1.17E-14	
Adjusted R-squared	0.972828	S.D. dependent var	6.518784	
S.E. of regression	1.074554	Akaike info criterion	3.019677	
Sum squared resid	143.1786	Schwarz criterion	3.130522	
Log likelihood	-189.7692	Hannan-Quinn criter.	3.064716	
F-statistic	1146.678	Durbin-Watson stat	1.656307	
Prob(F-statistic)	0.000000			

Dependent Variable: CRED Method: Least Squares Date: 03/10/23 Time: 09:51 Sample (adjusted): 2006M04 2016M12 Included observations: 129 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	31.13936	1.898327	16.40358	0.0000
NPL(-1)	-0.457424	0.385295	-1.187203	0.2374
IR(-1)	-0.737425	0.065386	-11.27798	0.0000
GDP(-1)	5.37E-06	1.24E-07	43.34862	0.0000
R-squared	0.952743	Mean dependent var	43.77713	
Adjusted R-squared	0.951609	S.D. dependent var	7.489106	
S.E. of regression	1.647447	Akaike info criterion	3.866848	
Sum squared resid	339.2604	Schwarz criterion	3.955524	
Log likelihood	-245.4117	Hannan-Quinn criter.	3.902879	
F-statistic	840.0429	Durbin-Watson stat	0.052351	
Prob(F-statistic)	0.000000			

Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 2 lags

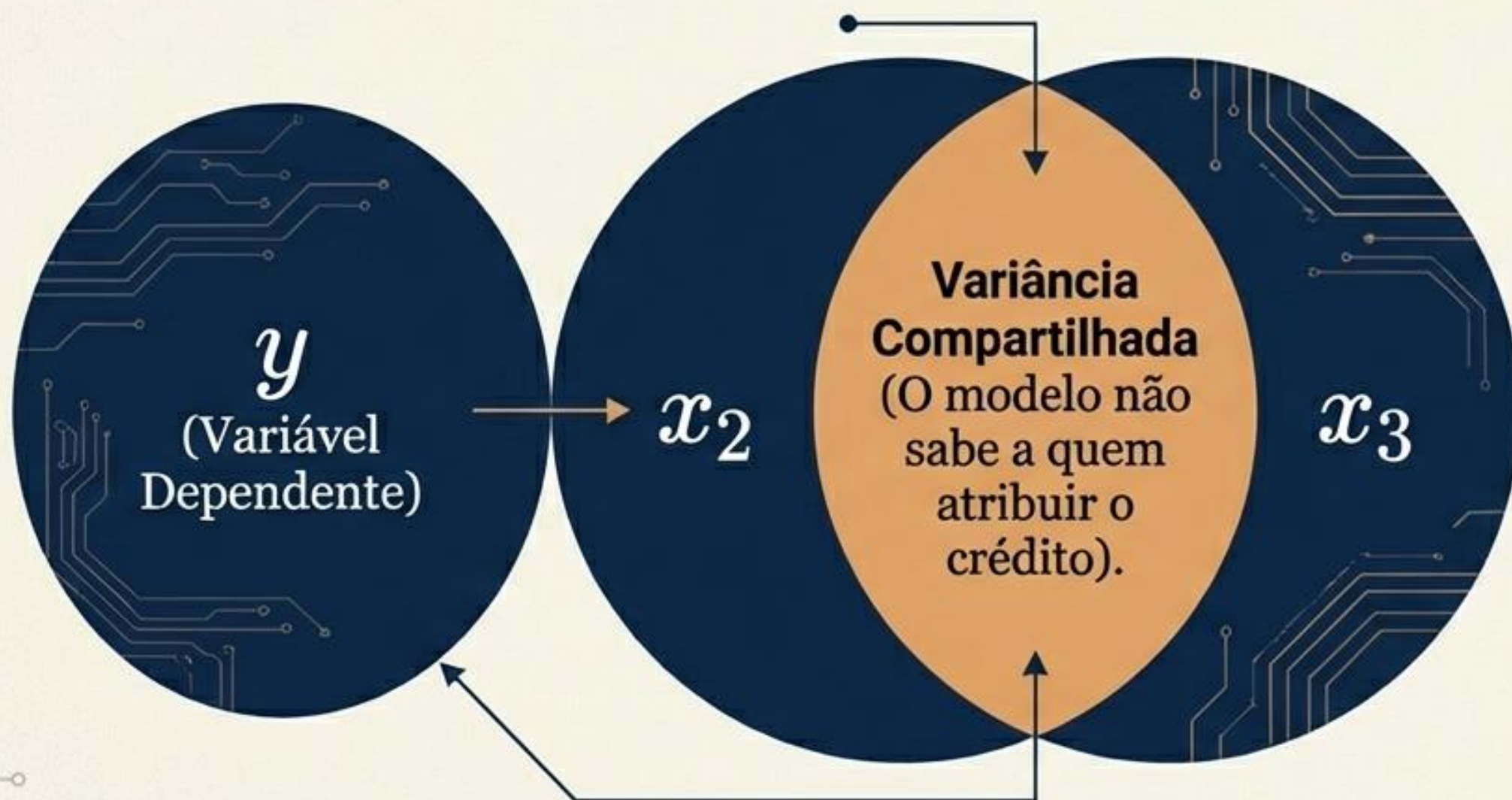
F-statistic	1295.993	Prob. F(2,123)	0.0000
Obs*R-squared	123.1558	Prob. Chi-Square(2)	0.0000

Test Equation:
Dependent Variable: RESID
Method: Least Squares
Date: 03/16/23 Time: 16:10
Sample: 2006M04 2016M12
Included observations: 129
Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.711192	0.461071	3.711343	0.0003
NPL(-1)	-0.334875	0.094872	-3.529755	0.0006
IR(-1)	-0.024107	0.014150	-1.703622	0.0910
GDP(-1)	-7.16E-08	2.80E-08	-2.561799	0.0116
RESID(-1)	1.115326	0.092969	11.99671	0.0000
RESID(-2)	-0.126290	0.095500	-1.322398	0.1885
R-squared	0.954696	Mean dependent var	-6.82E-15	
Adjusted R-squared	0.952854	S.D. dependent var	1.628027	
S.E. of regression	0.353495	Akaike info criterion	0.803498	
Sum squared resid	15.36988	Schwarz criterion	0.936512	
Log likelihood	-45.82562	Hannan-Quinn criter.	0.857544	
F-statistic	518.3972	Durbin-Watson stat	1.985095	
Prob(F-statistic)	0.000000			

Diagnóstico 3: Conflito de Informação (Multicolinearidade)

Quando as variáveis explicativas contam a mesma história.



Sintoma: R^2 alto, mas poucas variáveis individuais significativas (t-ratios baixos).

Métrica de Detecção: VIF (Fator de Inflação da Variância).



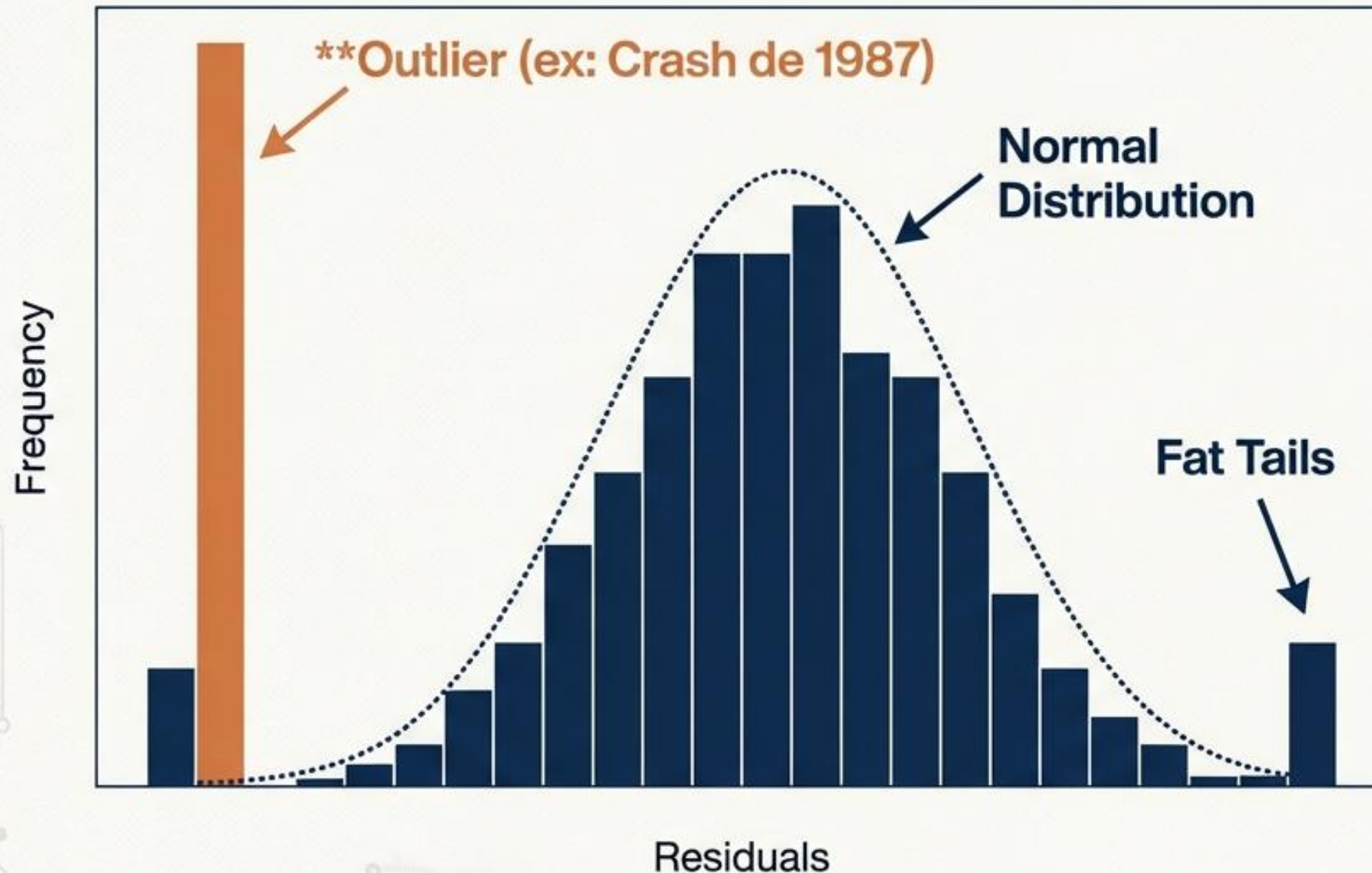
Regra de Bolso: $VIF > 10$ indica problema severo.



Solução: Remover uma das variáveis redundantes ou combiná-las em uma razão/ratio.



Diagnóstico 4: Normalidade e Eventos Extremos



Teste Formal:

Bera-Jarque (Baseado em Assimetria e Curtose).



Por que importa?

Fundamental para inferência em pequenas amostras. Eventos extremos violam a normalidade e distorcem o modelo.



Tratamento:

Uso de variáveis *Dummy* para controlar choques exógenos isolados.

6- Normalidades dos Resíduos

TESTE JARQUE-BERA

View/Residual Diagnostics/Histogram – Normality Test

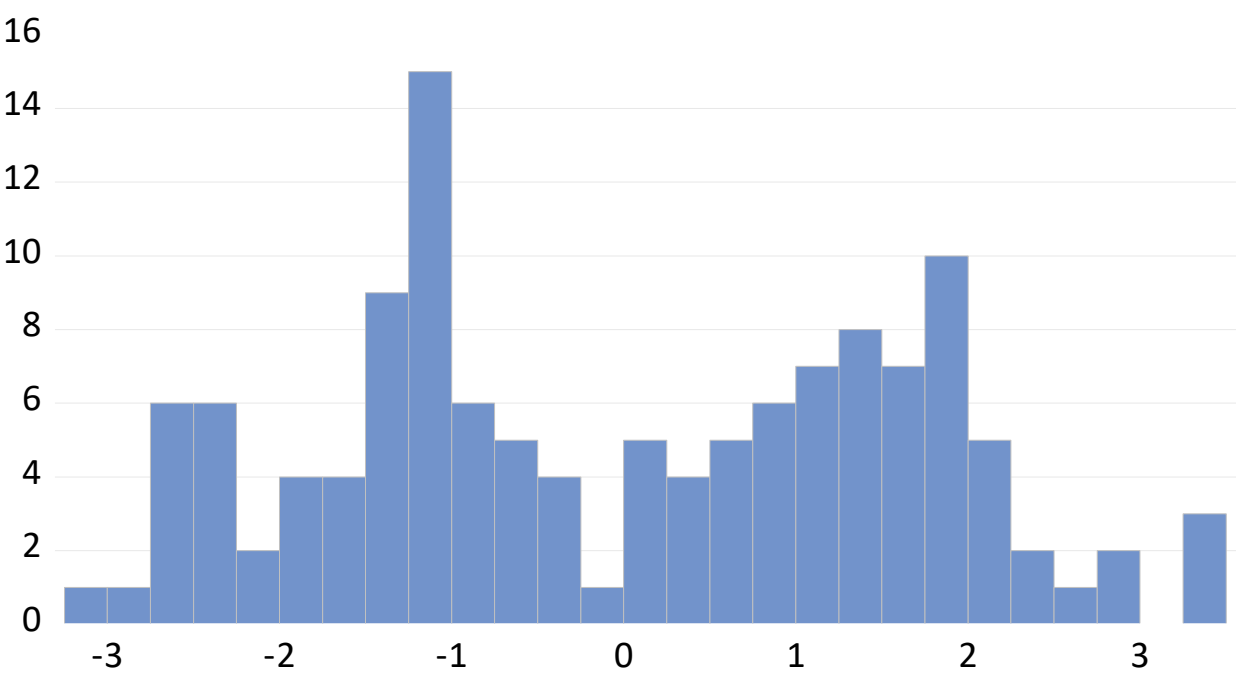
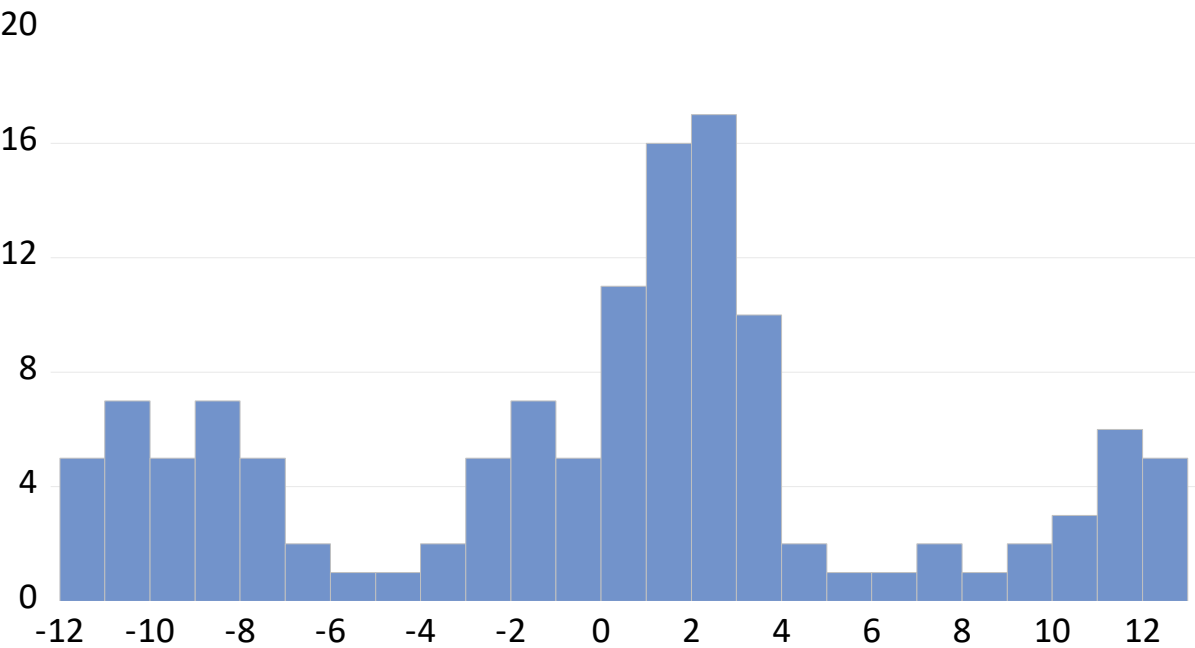
H0: Os resíduos são normalmente distribuídos

Dependent Variable: CRED
Method: Least Squares
Date: 03/10/23 Time: 09:47
Sample (adjusted): 2006M04 2016M12
Included observations: 129 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	81.49080	5.988192	13.60858	0.0000
NPL(-1)	-7.367373	1.398951	-5.266354	0.0000
IR(-1)	-1.079624	0.258865	-4.170611	0.0001
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Adjusted R-squared	0.230316	S.D. dependent var	7.489106	
S.E. of regression	6.570317	Akaike info criterion	6.625982	
Sum squared resid	5439.302	Schwarz criterion	6.692490	
Log likelihood	-424.3759	Hannan-Quinn criter.	6.653006	
F-statistic	20.15097	Durbin-Watson stat	0.017075	
Prob(F-statistic)	0.000000			

Dependent Variable: CRED
Method: Least Squares
Date: 03/10/23 Time: 09:51
Sample (adjusted): 2006M04 2016M12
Included observations: 129 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	31.13936	1.898327	16.40358	0.0000
NPL(-1)	-0.457424	0.385295	-1.187203	0.2374
IR(-1)	-0.737425	0.065386	-11.27798	0.0000
GDP(-1)	5.37E-06	1.24E-07	43.34862	0.0000
R-squared	0.952743	Mean dependent var	43.77713	
Adjusted R-squared	0.951609	S.D. dependent var	7.489106	
S.E. of regression	1.647447	Akaike info criterion	3.866848	
Sum squared resid	339.2604	Schwarz criterion	3.955524	
Log likelihood	-245.4117	Hannan-Quinn criter.	3.902879	
F-statistic	840.0429	Durbin-Watson stat	0.052351	
Prob(F-statistic)	0.000000			



E AÍ? O QUE FAZER?

REGRESSÃO ROBUSTA

Estimate/Options/HAC (Newey-West)

Dependent Variable: CRED
Method: Least Squares
Date: 03/10/23 Time: 09:47
Sample (adjusted): 2006M04 2016M12
Included observations: 129 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	81.49080	5.988192	13.60858	0.0000
NPL(-1)	-7.367373	1.398951	-5.266354	0.0000
IR(-1)	-1.079624	0.258865	-4.170611	0.0001
R-squared	0.242342	Mean dependent var	43.77713	
Adjusted R-squared	0.230316	S.D. dependent var	7.489106	
S.E. of regression	6.570317	Akaike info criterion	6.625982	
Sum squared resid	5439.302	Schwarz criterion	6.692490	
Log likelihood	-424.3759	Hannan-Quinn criter.	6.653006	
F-statistic	20.15097	Durbin-Watson stat	0.017075	
Prob(F-statistic)	0.000000			

Dependent Variable: CRED
Method: Least Squares
Date: 03/16/23 Time: 16:17
Sample (adjusted): 2006M04 2016M12
Included observations: 129 after adjustments
HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	81.49080	11.84987	6.876938	0.0000
NPL(-1)	-7.367373	2.335038	-3.155141	0.0020
IR(-1)	-1.079624	0.568525	-1.898991	0.0599
R-squared	0.242342	Mean dependent var	43.77713	
Adjusted R-squared	0.230316	S.D. dependent var	7.489106	
S.E. of regression	6.570317	Akaike info criterion	6.625982	
Sum squared resid	5439.302	Schwarz criterion	6.692490	
Log likelihood	-424.3759	Hannan-Quinn criter.	6.653006	
F-statistic	20.15097	Durbin-Watson stat	0.017075	
Prob(F-statistic)	0.000000	Wald F-statistic	5.135016	
Prob(Wald F-statistic)	0.007180			

E AÍ? O QUE FAZER?

REGRESSÃO ROBUSTA

Estimate/Options/HAC (Newey-West)

Dependent Variable: CRED

Method: Least Squares

Date: 03/10/23 Time: 09:51

Sample (adjusted): 2006M04 2016M12

Included observations: 129 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	31.13936	1.898327	16.40358	0.0000
NPL(-1)	-0.457424	0.385295	-1.187203	0.2374
IR(-1)	-0.737425	0.065386	-11.27798	0.0000
GDP(-1)	5.37E-06	1.24E-07	43.34862	0.0000

R-squared	0.952743	Mean dependent var	43.77713
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Sum squared resid	339.2604	Schwarz criterion	3.955524
Log likelihood	-245.4117	Hannan-Quinn criter.	3.902879
F-statistic	840.0429	Durbin-Watson stat	0.052351
Prob(F-statistic)	0.000000		

Dependent Variable: CRED

Method: Least Squares

Date: 03/16/23 Time: 16:20

Sample (adjusted): 2006M04 2016M12

Included observations: 129 after adjustments

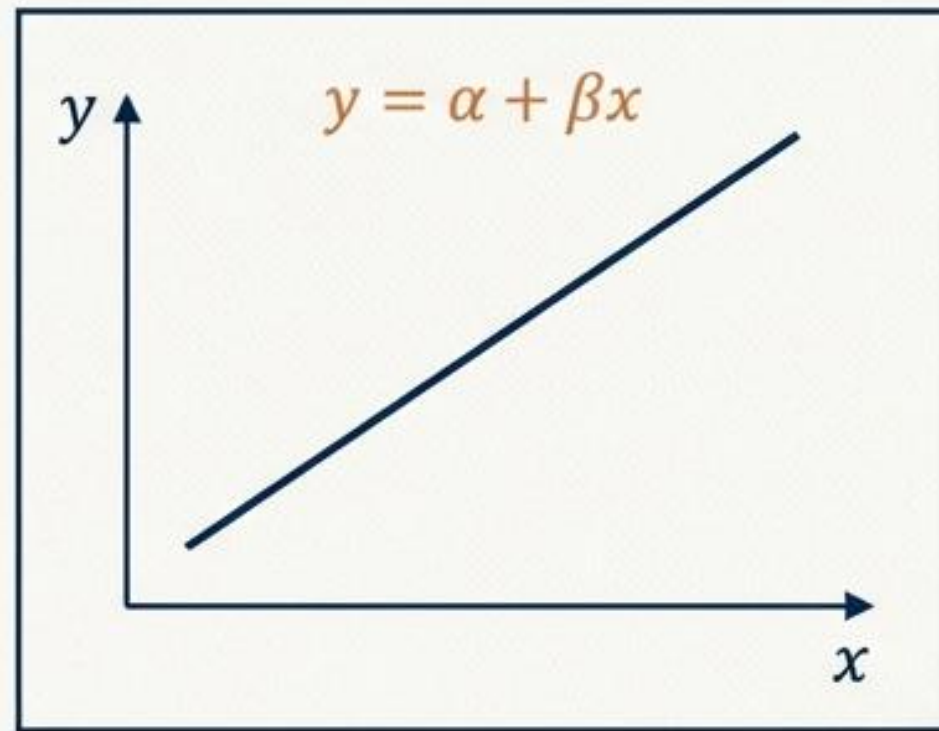
HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	31.13936	3.904468	7.975315	0.0000
NPL(-1)	-0.457424	0.821589	-0.556755	0.5787
IR(-1)	-0.737425	0.129788	-5.681774	0.0000
GDP(-1)	5.37E-06	2.89E-07	18.55133	0.0000

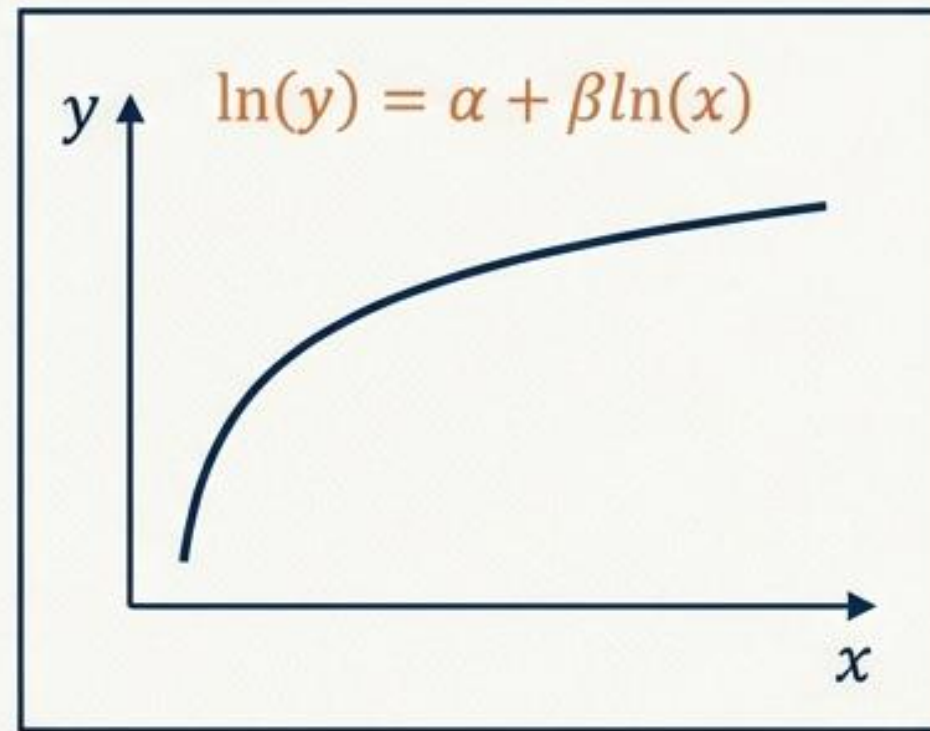
R-squared	0.952743	Mean dependent var	43.77713
Adjusted R-squared	0.951609	S.D. dependent var	7.489106
S.E. of regression	1.647447	Akaike info criterion	3.866848
Sum squared resid	339.2604	Schwarz criterion	3.955524
Log likelihood	-245.4117	Hannan-Quinn criter.	3.902879
F-statistic	840.0429	Durbin-Watson stat	0.052351
Prob(F-statistic)	0.000000	Wald F-statistic	160.7305
Prob(Wald F-statistic)	0.000000		

Diagnóstico 5: Especificação e Forma Funcional

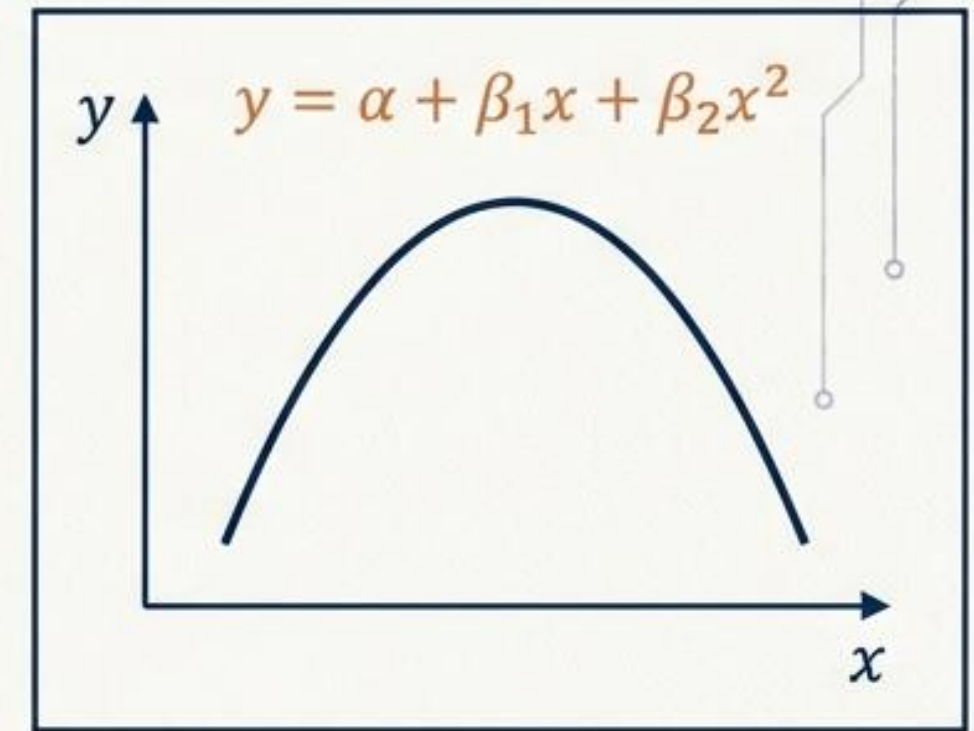
Estamos tentando medir uma curva com uma régua reta?



1. Linear



2. Log-Log (Elasticidade)



3. Quadrático



Teste RAMSEY RESET: Verifica se potências dos valores ajustados ($\hat{y}^2$, $\hat{y}^3$) explicam o erro remanescente.



Conselho: Escolha a forma baseada na teoria financeira, não apenas no ajuste dos dados.

DIAGNÓSTICO DE ESTABILIDADE

Ramsey RESET Test
Equation: OLS CRED 0
Omitted Variables: Squares of fitted values
Specification: CRED C NPL(-1) IR(-1)

	Value	df	Probability
t-statistic	0.512020	125	0.6095
F-statistic	0.262165	(1, 125)	0.6095
Likelihood ratio	0.270270	1	0.6032

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	11.38406	1	11.38406
Restricted SSR	5439.302	126	43.16906
Unrestricted SSR	5427.918	125	43.42334

LR test summary:

	Value
Restricted LogL	-424.3759
Unrestricted LogL	-424.2407

Unrestricted Test Equation:
Dependent Variable: CRED
Method: Least Squares
Date: 03/16/23 Time: 17:13
Sample: 2006M04 2016M12
Included observations: 129
HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	188.4210	354.7058	0.531204	0.5962
NPL(-1)	-20.50410	44.10658	-0.464876	0.6428
IR(-1)	-2.993322	6.316074	-0.473921	0.6364
FITTED^2	-0.020629	0.067618	-0.305081	0.7608

R-squared	0.243928	Mean dependent var	43.77713
Adjusted R-squared	0.225782	S.D. dependent var	7.489106
S.E. of regression	6.589639	Akaike info criterion	6.639391
Sum squared resid	5427.918	Schwarz criterion	6.728067
Log likelihood	-424.2407	Hannan-Quinn criter.	6.675422
F-statistic	13.44270	Durbin-Watson stat	0.018045
Prob(F-statistic)	0.000000	Wald F-statistic	3.842043
Prob(Wald F-statistic)	0.011359		

RAMSEY RESET TEST

View/Stability Diagnostics/Ramsey RESET Test

H0: O modelo não tem variáveis omitidas

Ramsey RESET Test
Equation: OLS CRED
Omitted Variables: Squares of fitted values
Specification: CRED C NPL(-1) IR(-1) GDP(-1)

	Value	df	Probability
t-statistic	12.32418	124	0.0000
F-statistic	151.8855	(1, 124)	0.0000
Likelihood ratio	103.1619	1	0.0000

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	186.7758	1	186.7758
Restricted SSR	339.2604	125	2.714083
Unrestricted SSR	152.4846	124	1.229714

LR test summary:

	Value
Restricted LogL	-245.4117
Unrestricted LogL	-193.8308

Unrestricted Test Equation:
Dependent Variable: CRED
Method: Least Squares
Date: 03/16/23 Time: 17:06
Sample: 2006M04 2016M12
Included observations: 129
HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	57.60783	5.106529	11.28121	0.0000
NPL(-1)	-1.857178	0.671197	-2.766964	0.0065
IR(-1)	-2.580535	0.278727	-9.258292	0.0000
GDP(-1)	2.15E-05	2.17E-06	9.885679	0.0000
FITTED^2	-0.034815	0.004628	-7.521858	0.0000

R-squared	0.978760	Mean dependent var	43.77713
Adjusted R-squared	0.978075	S.D. dependent var	7.489106
S.E. of regression	1.108925	Akaike info criterion	3.082648
Sum squared resid	152.4846	Schwarz criterion	3.193493
Log likelihood	-193.8308	Hannan-Quinn criter.	3.127686
F-statistic	1428.505	Durbin-Watson stat	0.104856
Prob(F-statistic)	0.000000	Wald F-statistic	316.6996
Prob(Wald F-statistic)	0.000000		

Modelagem Dinâmica: Adicionando “Memória”



Por que usar Lags (y_{t-1} , x_{t-1})?

- 🕒 **Inércia:** Mudanças econômicas e políticas demoram a surtir efeito.
- 📈 **Overreaction:** Mercados reagem demais e depois corrigem.
- 📊 **Benefício Técnico:** Frequentemente resolve problemas de Autocorrelação nos resíduos sem necessidade de GLS.



Aplicação

Estimate > Method:GMM > Instrument List (GDP(-1 TO -3) IR(-1 TO -3) > Number of Iterations:2

Teste J de Sobreidentificação:

H_0 : O modelo está adequado (p-valor deve ser maior que >0,10)

Quantidade máxima de instrumentos: Instrument Rank / Observations < 1

Dependent Variable: CRED				
Method: Least Squares				
Date: 03/30/23 Time: 09:52				
Sample: 2006M03 2016M12				
Included observations: 130				
HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
NPL	-0.406064	0.819566	-0.495463	0.6211
IR	-0.727469	0.131110	-5.548551	0.0000
GDP	5.45E-06	2.94E-07	18.55587	0.0000
C	30.32203	3.930944	7.713676	0.0000
R-squared	0.952310	Mean dependent var	43.65200	
Adjusted R-squared	0.951175	S.D. dependent var	7.595226	
S.E. of regression	1.678271	Akaike info criterion	3.903692	
Sum squared resid	354.8909	Schwarz criterion	3.991923	
Log likelihood	-249.7399	Hannan-Quinn criter.	3.939543	
F-statistic	838.6951	Durbin-Watson stat	0.050314	
Prob(F-statistic)	0.000000	Wald F-statistic	164.1089	
Prob(Wald F-statistic)	0.000000			

Equation Estimation

Specification

Options

Equation specification

Dependent followed by regressors, AR and PDL terms, OR an explicit equation.

cred npl ir gdp c

Instrument list

gdp(-1 to -3) ir(-1 to -3)

☒ Include a constant

Estimation weighting matrix

HAC (Newey-West)

HAC options

Weight updating

N-Step Iterative

Number of iterations: 2

Estimation settings

Method: GMM - Generalized Method of Moments

Sample: 2006m03 2016m12

OK

Cancelar

Dependent Variable: CRED				
Method: Generalized Method of Moments				
Date: 03/30/23 Time: 10:00				
Sample (adjusted): 2006M06 2016M12				
Included observations: 127 after adjustments				
Linear estimation with 2 weight updates				
Estimation weighting matrix: HAC (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)				
Standard errors & covariance computed using HAC weighting matrix (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)				
Instrument specification: GDP(-1 TO -3) IR(-1 TO -3)				
Constant added to instrument list				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
NPL	-2.689891	1.214155	-2.215442	0.0286
IR	-0.767008	0.131472	-5.833984	0.0000
GDP	5.29E-06	2.78E-07	19.02148	0.0000
C	39.05888	5.310840	7.354558	0.0000
R-squared	0.930520	Mean dependent var	44.02252	
Adjusted R-squared	0.928825	S.D. dependent var	7.284314	
S.E. of regression	1.943353	Sum squared resid	464.5242	
Durbin-Watson stat	0.044390	J-statistic	2.415782	
Instrument rank	7	Prob(J-statistic)	0.490704	

O Checklist do Analista (Resumo Executivo)

Problema (Doença)	Sintoma Visual	Teste Formal	Solução (Cura)
Heterocedasticidade	Formato de Cone	Teste de White	Erros Robustos / GLS / Log
Autocorrelação	Ondas ou Dentes	Breusch-Godfrey / DW	Newey-West / Lags
Não-Normalidade	Caudas Gordas / Outliers	Bera-Jarque	Dummies / Transformação
Instabilidade	Mudança de Inclinação	Teste de Chow	Dividir Amostra
Multicolinearidade	R^2 alto, t-ratio baixo	VIF > 10	Remover variável / Razão

Conclusão: A Arte da Modelagem

“Não torture os dados até que eles confessem.”

Testes de diagnóstico não são burocracia; são gestão de risco financeiro.

Um modelo com premissas violadas é um modelo enganoso.

Regra de Ouro: Prefira sempre a robustez teórica e a simplicidade sobre a complexidade cega.

Antes de apresentar qualquer modelo para a diretoria, execute o 'Check-up de Saúde'.

